

Tracking Training in Ultrarunning

I killed a lot of trees when I first started coaching. Throughout the course of a workday, I received numerous faxes and email attachments of training files from my athletes. I printed out every single one of them, and then I'd pore over each file, examining the critical data. Yellow, orange, and pink highlights would adorn the sheets, denoting intervals, critical heart rate values, and anything else I found significant. I'd fill the margins with notes from conversations and debriefs on each workout from the athlete. Finally, I'd place the sheets of paper in individually labeled folders, one for each athlete.

At the conclusion of a training phase, I'd spread the contents of an athlete's folder out on a table. My preferred canvas for this exercise was a large, round table in our conference room, but that seldom gave me enough space. Papers were everywhere—on the table, floor, and chairs and in the adjoining rooms; no horizontal surface was safe. Once the documents were strewn about to my liking, I'd try to make sense of it all. Was the athlete improving? Were they hitting the correct intensities? Was it time for rest? The colors, highlights, notes, and scribbles all provided clues; the conclusions, however, I had to determine myself. As rudimentary as this system was, it did its job, and I could get a good fix on what was going on with the athlete. It was not perfect, but it was all I had.

Over time, training-related technology and software has improved and is now quite sophisticated. Sheets of paper collected in file folders gave way to electronic storage and access. The device manufacturers, motivated to differentiate their products from the rest, all created ways of organizing and making sense of the information collected from their widgets. Each system has its own spin. Some are very good at taking individual files and breaking them down. Others are better at aggregating many files to identify trends. All have been helpful to coaches trying to understand it all.

As heart rate training gave way to power-based training (in cycling) and GPS-based training (in running), some of our previously unidentified training flaws started to show. We could now see pace and power decoupled from heart rate (renowned endurance coach Joe Friel developed terms to describe this phenomenon: “aerobic decoupling” and “efficiency factor”). We could quantify and aggregate training load and see how various loads affected performance. Myriad other insights ensued, none of which would have been possible from staring at 8½ × 11-inch sheets of paper, no matter how much highlighter I used.

As a coach working with endurance athletes, I took to this evolution, which helped me gain greater insight into how my athletes were training. In road running, triathlon, and cycling, the data acquired in the field could be meaningfully interpreted in order to analyze and adapt training for athletes. I had tangible quantifications of workload, and improvements could be easily identified. I wish I could say the same kind of data existed for trail ultrarunners.

In ultrarunning, particularly trail ultrarunning, the tangible quantification of workload and work rate that exists in road running and cycling becomes inconsistent and blurry. Improvement markers are also difficult to come by, particularly in mountainous areas where the properties of the trail surface can change from one run to the next or even during a single run.

Nonetheless, when I started working with ultrarunners, I realized I needed to solve the same problems I'd faced with other athletes. I needed a way to quantify workloads and work rates and to compare one workout to the next. I needed a way to determine if an athlete had improved. Harking back to my earlier experiences with folders and highlighted sheets of paper, I hacked together a way to use the available tools and data to better coach ultrarunners. Is it perfect? Hell no. Does it do the job? You bet. Technology and software will continue to evolve, but in the meantime, people still have to run and train, and I—and you—have to figure out ways to make sense of it all.

Tools to Track Your Training

GPS monitors are the primary tracking devices for recording day-to-day ultrarunning performance. Several manufacturers produce them, and ultrarunners have to pick the features and price point that best suit their needs. This technology is constantly evolving, so rather than focus on specific brands and the differences between them, I will focus on the key features to look for when selecting and using a device to track your training.

GPS ACCURACY

Much of what you can do to train for the specificity of any particular ultra is rooted in the exercise of matching the grades and surfaces you train on to what you will experience on race day. This being the case, GPS and altitude accuracy are important considerations in choosing a watch. The quality of GPS data is fundamentally determined by the strength of the satellite signal, how many satellites are acquired, the position of the antenna within the watch, and how frequently the satellite information is collected. Most of the traditional watch manufacturers do a great job with this aspect, and there are few notable differences among high-quality watches.

BATTERY LIFE

Battery life on many GPS watches hinges on the quality and size of the battery and the amount and quality of data samples collected. Many of the device manufacturers are becoming wise to the fact that athletes are going longer, so they are engineering their watches and batteries to last as long as possible and providing a reasonable footprint for the watch to wrap around your wrist. Battery technology is improving all the time, stretching the life span of what a watch can record in a single session. Some watch manufacturers are even embedding small solar panels into the watch as a means of extending battery life.

Taking battery life a step further, device manufacturers also provide options for athletes to manipulate the GPS recording variables of the watch. The user can opt to save battery life by recording data at less frequent intervals or by manipulating how the GPS signal is acquired. The watch then uses non-GPS movement data to fill in the gaps and determine speed when GPS position is not being recorded. Some GPS devices come with this as the only recording option to minimize the watch's physical size and bring it down to a less expensive price point. It is a classic compromise that prioritizes the battery life (and sometimes price point) over the accuracy of the data collected.

As another approach, some manufacturers give athletes the ability to improve GPS quality by utilizing GLONASS and GALILEO satellite capabilities and other GPS signal-strengthening tools. This decreases the battery life for any one particular workout but improves the accuracy of the data collected.

As a coach, I prefer to have the highest-quality data possible. In analyzing files, it makes a difference looking at information that was collected every second vs information that was collected every few seconds, particularly when evaluating intervals that are just a few minutes in length. From a practical standpoint, interval recording options of one second are necessary for any trail-running GPS watch. Furthermore, because trail running is done where terrain and trees can interfere with satellite signals, a premium must be placed on acquiring the most precise position possible at all times. Therefore, while the option of sparing battery life at the expense of lower-quality data is a good one (particularly for longer races), in day-to-day training it is best to stick to the highest-quality data and most frequent data collection rate possible.

Logging Your Training

After you choose the right device to monitor and track your training, the next step is to home in on the software that is going to harness the data and give you actionable information you can use to better guide and focus your training. As a coach, I have thousands of hours of practice aggregating and analyzing training information. I don't ask my athletes, and I would not ask you, to delve that deeply into the data. However, with a few key insights into how concepts like total workload, work rate, fitness, and fatigue work, you can use the information collected by your GPS watch, as well as personal feedback, to better monitor your training.

THE TOOLS OF STRAVA AND TRAININGPEAKS

While each device manufacturer (Garmin, Suunto, COROS, etc.) has developed its own software to harness and present data, third-party developers have spent the last several years showing us that they can do a better job in this role. Garmin, for example, makes numerous GPS devices (and even cameras) for use in running, golf, boating, driving, aviation, and other activities. Suunto, Polar, and other GPS device manufacturers similarly make a range of different products. All of these manufacturers are experts in making devices that can capture information on a miniature computer (like, in the case of running, a watch). Meanwhile, software-specific companies focus solely on taking the data acquired from a training device and presenting actionable information. They are experts in taking information from the tiny computers and making sense out of it for athletes and coaches. So, it is not surprising that the better platforms for monitoring and tracking training come from the software-specific manufacturers. In my view, Strava and TrainingPeaks are currently the go-to platforms for endurance athletes.

COMPARING WORKLOAD AND WORK RATE

Determining total workload (kilojoules) and work rate (power) is one of the long-standing challenges in coaching trail ultrarunners. While volume quantifies the amount of training (in time or miles), and intensity describes the effort of training, total workload and work rate are more useful in determining how much training stress an athlete incurs.

According to simple Newtonian physics, work is a product of force and distance. A bigger force applied over a longer distance leads to more work done. In level running, the amount of work performed is a function of your weight (which, as was discussed in Chapter 5, influences ground reaction forces) and how far you ran (distance). Therefore, for level running, workload can be well represented by how many miles you ran and how much you weigh. A heavier runner will have a higher total workload than a lighter runner who covers the same number of miles.

Work rate is the total amount of work done divided by the time it took to perform it. On level ground, running a mile takes approximately the same amount of work regardless of how long it takes to run it. You can see this manifested in caloric expenditure equations (total calories = 1 calorie per kilogram per kilometer); caloric expenditure is a function of distance and weight, not speed. Your work rate, however, is dependent on speed. The faster you run any given distance, the higher your work rate. For *flat* running, work rate can be represented by pace and weight. Similarly, a heavier runner will have a higher work rate than a lighter runner at the same speed.

For a runner who is training primarily on *flat* terrain, workload can be represented by miles, and work rate by pace. These two values can even be compared among different runners who have the same mass. In cycling, work rate and workload can actually be measured by devices called power meters. At the time of this writing, nothing analogous is available in trail running. (Sit tight before you yell at me about running power meters like Stryd,

RunScribe, and Garmin.) The differences in trail surface and elevation gain and loss make capturing workload and work rate problematic at best. There are tools out there that trail runners and ultrarunners can use, but to properly evaluate training, it is important to know the basis for and limitations of such tools.

NORMALIZED GRADED PACE AND GRADE ADJUSTED PACE

Both Strava and TrainingPeaks have developed algorithms to convert running on uphill and downhill to fl at, level running (Table 6.1). TrainingPeaks' Normalized Graded Pace (NGP) and Strava's Grade Adjusted Pace (GAP) take your running and "grade" it as if you were running on fl at, level ground. For example, if you are running up a 6 percent climb at a 12:00 min/mi pace, your NGP or GAP would be 9:21. In other words, running 12:00 min/mi on a 6 percent uphill is comparable to running 9:21 min/mi on the fl ats. This gives athletes the ability to compare paces on different climbs and descents with the equivalent pace for flat, level running.

PACE PER MILE	GRADE		-10%		-6%		-2%		0%		2%		6%		10%	
			NGP		GAP		NGP		GAP		NGP		GAP		NGP	
8:00			15:18	11:27	11:13	10:05	8:51	8:40	8:00	8:00	7:19	7:22	6:14	6:16	5:26	5:21
10:00			19:07	14:19	14:00	12:37	11:04	10:50	10:00	10:00	9:09	9:13	7:48	7:50	6:47	6:42
12:00			22:56	17:11	16:48	15:08	13:17	13:00	12:00	12:00	10:59	11:04	9:21	9:24	8:08	8:02

TABLE 6.1 NGP and GAP for different grades and paces.

TrainingPeaks' NGP takes the paced algorithm a step further (see what I did there?) and "normalizes" the pace to its physiological equivalent had the runner perfectly paced the entire run. For example, if you do a workout comprising of six sets of three minutes hard, three minutes easy, where the hard parts are at six-minute pace and the easy parts are at ten-minute pace, your average pace for that workout will be just under seven minutes per mile for thirty-three total minutes. But, the workout will certainly feel harder and have more physiological impact than thirty-three minutes at seven-minute pace. So, TrainingPeaks' NGP "normalizes" undulating paces seen in training and racing and calculates a physiological equivalent.

Both of these algorithms do a decent job of comparing the respective paces (and therefore work rates) of climbing and flat, level running when performed on similar surfaces and at normal gradients. You can go out and do intervals on flat sections and on climbs, compare the efforts, and determine which effort was harder or easier. However, there are two glaring flaws in utilizing these algorithms for trail runners.

Flaw 1: When the Surface Is Different

Neither GAP nor NGP has the ability to account for the difference in work associated with running on different surfaces. You intuitively know that running through sand is more difficult than running on a track. Similarly, running over technical terrain requires more effort than running over smooth terrain. However, the raw paces and calculated GAP and NGP will not account for the differences between those surfaces.

Flaw 2: Descents

The calculations for GAP and NGP use the difference in energy/oxygen cost between uphill, downhill, and level running to arrive at the equivalent pace for level terrain. While on flat ground, your pace is directly related to work rate. Your cardiovascular system has to work harder in order to go faster, which means a higher pace requires a higher work rate. But remember that for downhill running in particular, energy cost tells only part of the story. For downhill running, other factors outside of the energetic cost combine to significantly affect the overall stress. These include changes in foot speed, coordination, and musculoskeletal stress, all of which are different in downhill running than in level or uphill running. Neither GAP nor NGP takes these additional stresses into account, and as a result those two algorithms underestimate the overall stress of downhill running. In a single or shorter run, the misestimation may not be a big issue. For ultrarunners, however, it represents a greater flaw as you try to sum up cumulative training stress during a very long run or over a longer period of training.

Total Training Stress

Total workload and work rate can be used to quantify the amount of training stress for any particular run. This is important because it gives an apples-to-apples comparison of how stressful different runs were. Hilly, undulating runs with wild differences in pace can be compared with flat runs with little deviation in pace on the basis of their total amount of stress. Similarly, interval workouts can be compared to EnduranceRuns. In TrainingPeaks, these values are referred to as Training Stress Scores (TSS), of which there are three flavors depending on how you want to delineate intensity (either by power denoted by TSS, pace denoted by rTSS, or heart rate denoted by hrTSS). Strava's version of this is called Relative Effort. All of these scoring systems are trying to do the same thing, albeit by different methods.

On the TrainingPeaks platform, training stress is calibrated using an algorithm anchored to a sixty-minute run at lactate threshold being equal to one hundred points. Recovery runs will be less than 100 points, and longer endurance runs will typically be more than one hundred points. This gives you a basis to determine how stressful a short, high-intensity run is compared with a longer, lower-intensity run (see Table 6.3 for examples). Additional formulas can utilize pace (in the case of rTSS), heart rate (in the case of hrTSS), or power (in the case of TSS) as the intensity determinant in order to let the user decide which flavor to use for a particular workout. The reason TrainingPeaks uses this 100-point calibration system is that one hour of running at lactate threshold intensity (what they term Functional Threshold Pace) is about the maximum that most normal people can sustain. (Elite athletes can sustain more than an hour at threshold and beginner athletes less, so they split the difference.) The effort would essentially be an all-out time trial. Therefore, when a coach or athlete looks at a particular run, using 100 points as an anchor for an extremely difficult, high-intensity effort, the score for the particular run is immediately put into a broad context.

PLATFORM	NAME	INTENSITY DETERMINANT	METHOD OF SCORING
Strava	Relative Effort™	Heart rate	Threshold heart rate, duration of run, time spent at different intensities
TrainingPeaks	TSS™	Running Power	Threshold power, duration of run, NGP for run
TrainingPeaks	rTSS™	NGP	Threshold pace, duration of run, NGP for run
TrainingPeaks	hrTSS™	Heart rate	Threshold heart rate, duration of run, NGP for run

TABLE 6.2 Training stress scoring systems.

On Strava’s platform, Relative Effort was inspired by Dr. Eric Bannister’s TRIMP (Training Impulse) system (Bannister and Calvert 1980). It uses heart rate and the time spent across different intensities to score a workout.

If you are caught up in the weeds right now, fret not. Here’s the bottom line: TSS/rTSS/hrTSS and Relative Effort use your lactate threshold intensity (defined by running power, pace, or heart rate) and the time you spend at your relative intensities (endurance, lactate threshold, VO2 max, etc.) to score each workout using a point-based system. The higher the score, the more stressful the workout.

TYPE OF RUN	TSS/rTSS/hrTSS POINTS
60-min RecoveryRun	50-80
90-min EnduranceRun with 3 x 10 min TempoRun	100-150
90-min EnduranceRun with 6 x 3 min RunningIntervals	100-150
2.5-hr EnduranceRun	150-200
50-mile race	400-600

TABLE 6.3 Training stress scoring systems.

There are ton of caveats to these scoring-based systems. Before we get to those, though, let's dive into how to leverage these training load metrics a bit more.

AGGREGATING TOTAL TRAINING STRESS

Within TrainingPeaks, your TSS/rTSS/hrTSS can be aggregated and trended over time. This provides one of the more valuable pieces of feedback when analyzing training load. Acute Training Load (ATL) and Chronic Training Load (CTL) provide snapshots of the long-term (more than seven days) aggregate training load.

Acute Training Load is a seven-day rolling weighted average of the TSS/rTSS/hrTSS for each particular day. Chronic Training Load is the same thing but over a forty-two-day period. (The time frames for CTL and ATL can be customized.) Over time, these numbers offer a big-picture view of historical training that provides clues as to when an athlete is the most fit (highest CTL), most fatigued (highest ATL), and most ready for performance (highest Training Stress Balance, or TSB, which is the difference between CTL and ATL). TrainingPeaks provides useful charting capabilities of these metrics that athletes can use simply by uploading their training files on a day-to-day basis.

[Figure 6.1 is an analysis of an athlete training for the Western States 100. You can see the athlete's CTL was highest just before Western States, indicating that they were at peak fitness for that race. You can also see the consistency with which the athlete built their training load over the course of the season, as indicated by the ramping gray area.](#)

Strava, Garmin, and other companies offer similar, albeit more rudimentary, tools to aggregate training stress. Strava's version, for example, is called "Fitness and Freshness." If you are interested in tracking your training load and gleaning valuable insights, these tools are not quite up to the task, and athletes are better served utilizing TrainingPeaks' ATL/CTL/TSB system for analysis.

BEWARE THE FLAWS

Unfortunately, the flaws with TSS/rTSS/hrTSS, ATL, and CTL mirror those with NGP (which is unsurprising given that NGP is used in the equations). Changes in surface and large amounts of descending make it difficult to compare one run with the next. However, if your training generally contains the same trail surface and amount of climbing and descending from day to day, rTSS, ATL, and CTL can give you a ballpark idea of how hard or easy one run is compared with the next and how these training loads stack up over time.

Also bear in mind that point for point within any training scoring system, points are physically more difficult to a runner the further they are into a run. In other words, fifty TSS points (equivalent to a short RecoveryRun) when accumulated at the beginning of a one-hour run are not nearly as difficult as when accumulated at the end of a six-hour run, yet they will be calculated at the same Training Stress. In addition, training scoring systems are essentially combining stress from endurance and intensity into one metric, which is inherently problematic. For example, a sixty-minute TempoRun (which is essentially an all-out effort) will accumulate approximately the same number of TSS or Relative Effort points as a two-and-a-half-hour EnduranceRun, although they have markedly different effects on the body and require different amounts of recovery (more recovery will be needed after a sixty-minute TempoRun).

MONITORING IMPROVEMENT

How do you know you if are becoming more fit? In the road-running world, your day-to-day paces and workouts can provide answers to this question. In trail running, the process is similar, though the answer requires more investigation, particularly if you are doing your specific interval work on trails (as you should be).

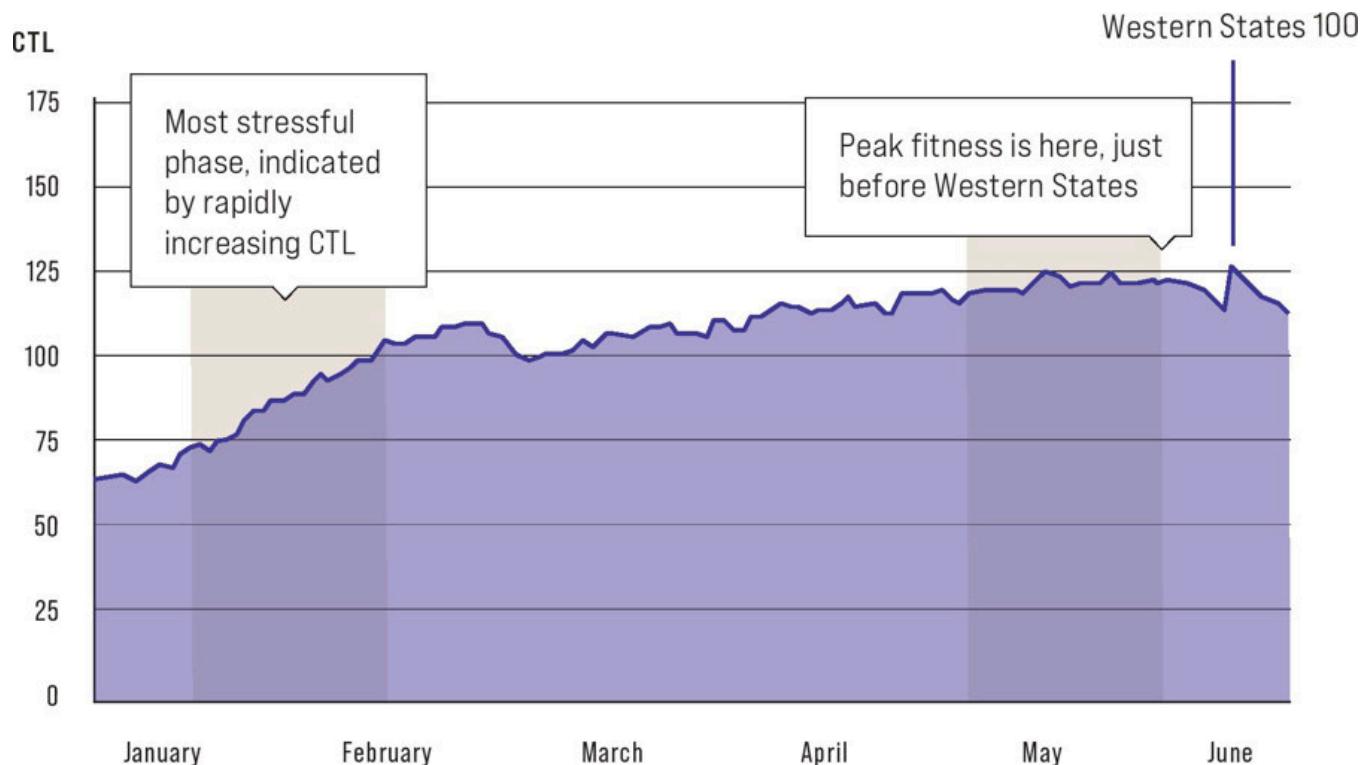


FIGURE 6.1 Analysis of an athlete's training for Western States. The CTL (blue-shaded area) is highest just before the Western States 100. This indicates that the athlete was most fit just before the race. The CTL also ramps up fastest during the tempo phases, indicating that they are generally the most stressful phases.

Both Strava and TrainingPeaks provide tools to help you understand how your fitness is trending. These tools do not provide spotlight answers that turn green when things are good, red when they are bad, and yellow for somewhere in between. Rather, they provide general trends you can interpret to see if your training is on the right track and if you are making improvements over a period of weeks or months.

Strava Segments

One of the great and convenient features of Strava is segment tracking. Strava segments are marked-out sections of road and trail, created by users. You can even create your own segments for sections of trail you commonly use for specific interval work. Every time you run across that section of trail, your segment time is recorded for everyone to see (you can change this setting if you wish to keep your run or your segment private). Results are ranked and stacked on a leaderboard, and you can see where your time fits in with the rest of the pack or in relation to your previous runs. While you should not try to set a PR every time, the general trend line can provide clues to how your fitness is trending, particularly if the segment of trail is one you use frequently for training (Figure 6.2). The great thing about these segments (other than being addictive) is that they are automatically

calculated and tabulated once you have uploaded your run—no need to print out pages and mark them up with a highlighter.

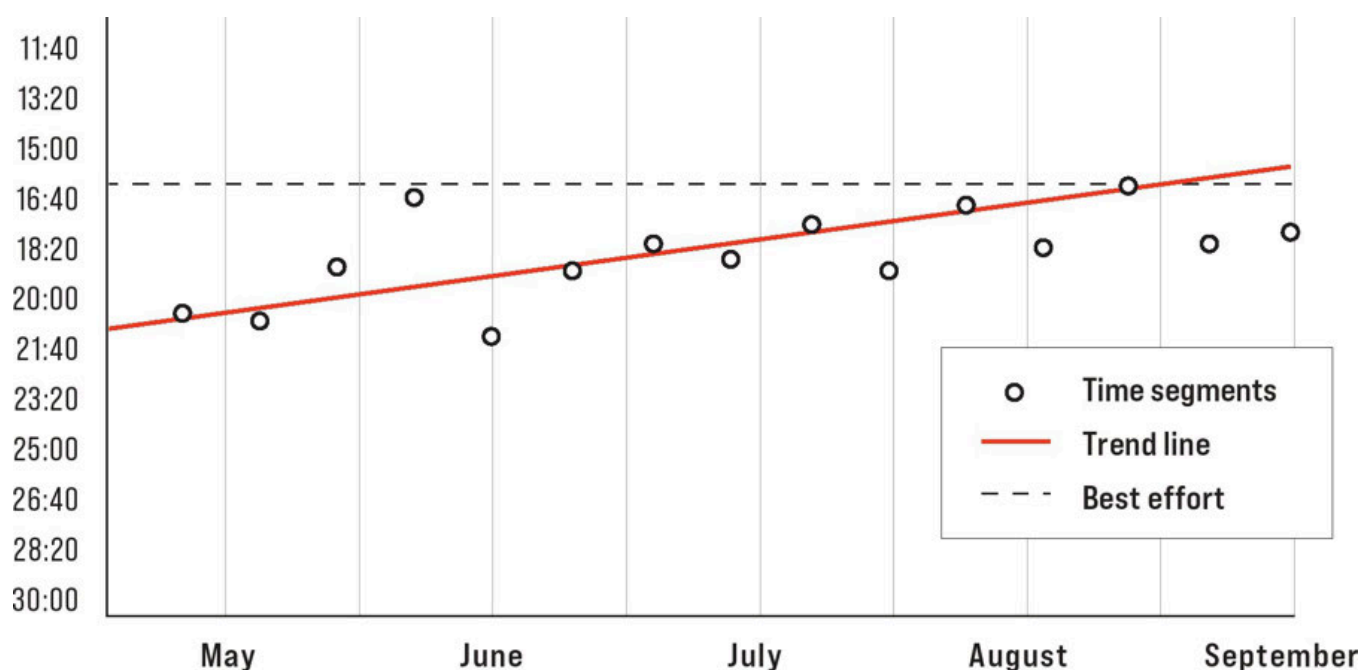


FIGURE 6.2 Strava segments with the trend line generally getting better over time.

NGP Between Efforts

TrainingPeaks provides fitness tracking via the capability to break down files and create laps once the file is uploaded. In this way, after doing specific workouts, you can come back to the file and analyze the NGP of specific segments. You can use these data to gauge the quality of your workouts from interval to interval (Figure 6.4) and to compare intervals across several days or weeks.

Measuring and Personalizing Training Intensity

If you are going to use interval training to accumulate time at intensity and target specific areas of your fitness, you need a way to figure out how hard you are working. In some sports like endurance running and swimming, this is simple. Ultrarunners don't have it so easy. For a long time, runners have tried to use heart rate to gauge intensity, creating intensity ranges based on percentages of lactate threshold heart rate or the average heart rate recorded during a 5K time trial. Others have used pace ranges based on time trials or goal race paces, or a combination of heart rate and pace ranges. Prescribing intensity based on either heart rate or pace is notoriously difficult in ultrarunning, and after trying all manner of methods, I found the greatest success in a remarkably simple, nontechnical, yet scientifically accurate method: rating of perceived exertion (RPE).

Why Heart Rate Is Not a Good Training Tool for Determining Intensity

The heart rate value you see on a watch is a measurement of your body's response to exercise. It's not a direct measure of the work being done; instead, the work is being done primarily by muscles, which in turn demand more oxygen from the cardiovascular system. Because that oxygen is delivered via red blood cells, heart rate increases in order to pump more oxygen-rich blood to the muscles as their demand for oxygen rises. It's an indirect observation of what's happening at the muscular level, but in the absence of a direct way to measure workload, heart rate can provide valuable information. Research has shown conclusively that there's a strong correlation between heart rate response and changes in an athlete's workload, and that research allowed sports scientists and coaches to start creating heart rate training zones back in the 1980s. However, as sports science has evolved over the past thirty-plus years, we have learned that many factors affect an athlete's heart rate, and those factors reveal that heart rate response is not reliable or predictable enough to be an effective training tool.

FACTORS AFFECTING HEART RATE

Core Temperature

As your core temperature increases, heart rate at a given exercise intensity will increase. Your circulatory system carries heat from your core to your extremities to aid with conductive and radiant cooling. Because part of your blood flow is redirected in this manner, your heart has to work harder (pump faster) to deliver the same amount of oxygen to the muscles.

Caffeine and Other Stimulants

When you consume caffeine, either from your morning cup of coffee or from a caffeinated gel during a training session or race, your heart rate increases.

Excitation/Nervousness

A race is an exciting event (or at least it should be if you have the right emotional engagement), and that causes a release of epinephrine and norepinephrine, which increases your heart rate. Other emotional responses, including frustration, anger, and anxiety, can also affect heart rate.

Hydration Status

Although hydration-based changes in heart rate are often concurrent with impacts from core temperature, your heart rate can increase from dehydration with or without a rise in core temperature. As your blood volume diminishes, your heart needs to beat faster to deliver the same amount of oxygen per minute.

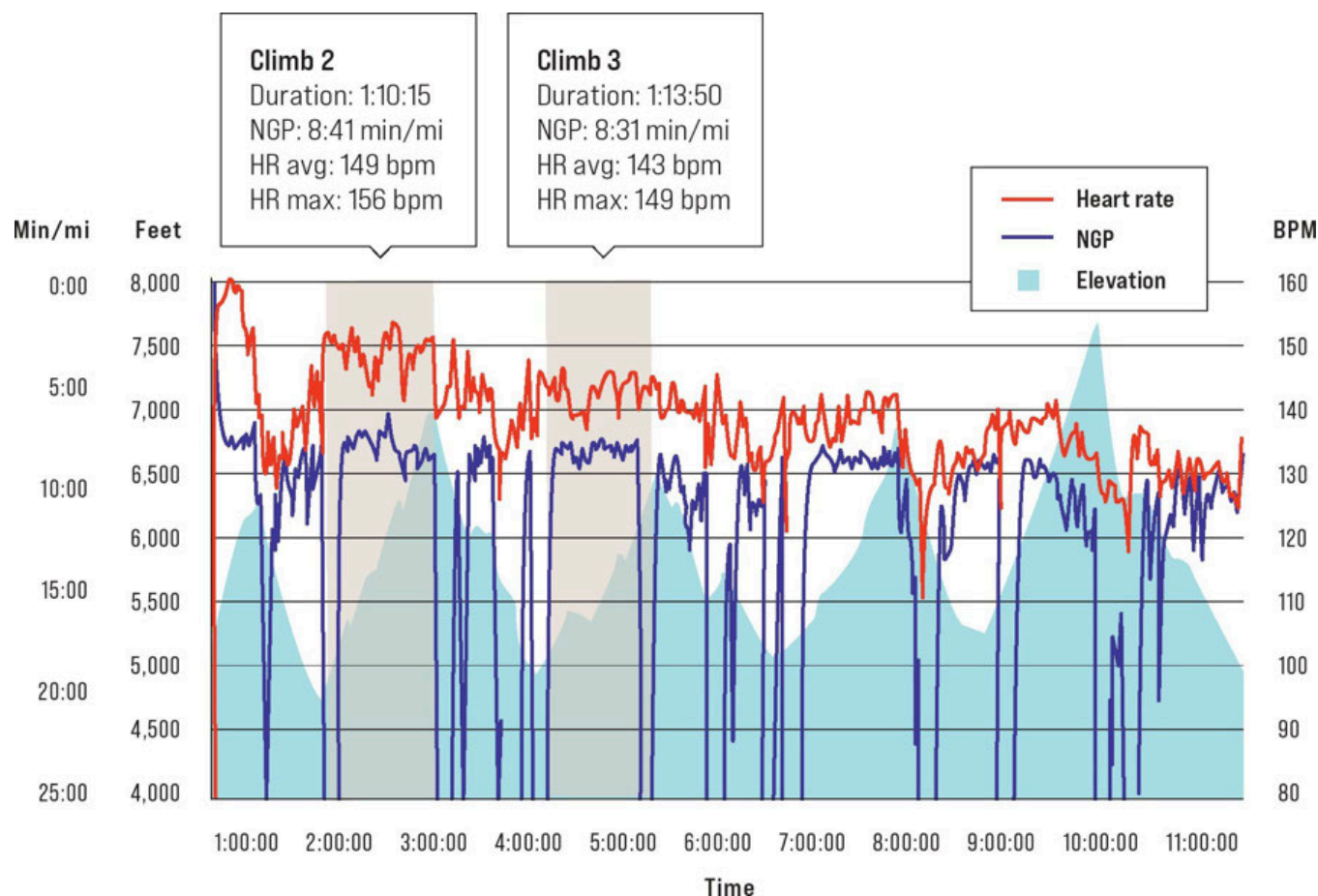


FIGURE 6.3 Example of how fatigue affects heart rate. Heart rate (red line) starts high due to freshness and then drops as fatigue sets in, even though NGP remains roughly the same for the first three climbs.

Altitude

Most athletes train within a small altitude range in their local area, but goal races may feature dramatically different elevation profiles. Heart rate and respiration rate increase at altitude, starting at about 5,000 feet above sea level. This occurs because the reduced partial pressure of oxygen in the air you're breathing means there are fewer oxygen molecules in each lungful of air.

Fatigue

While many of the factors that impact heart rate act to increase it, fatigue often suppresses it. When you are fatigued, your heart rate response to increasing energy demand is slower and blunted. A tired athlete will see heart rate climb more slowly at the beginning of an interval or hard effort and will struggle to achieve the heart rate normally associated with a given intensity level.

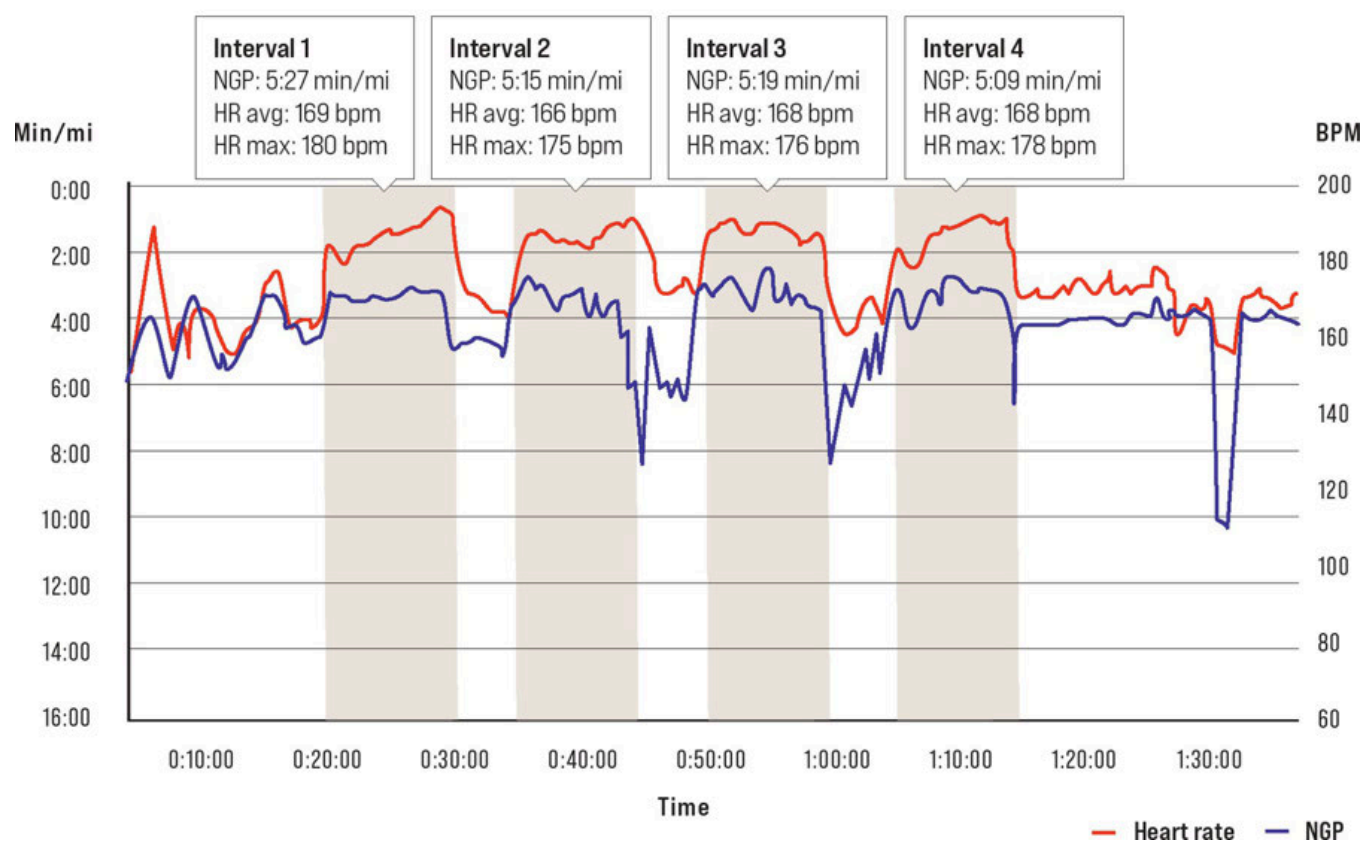
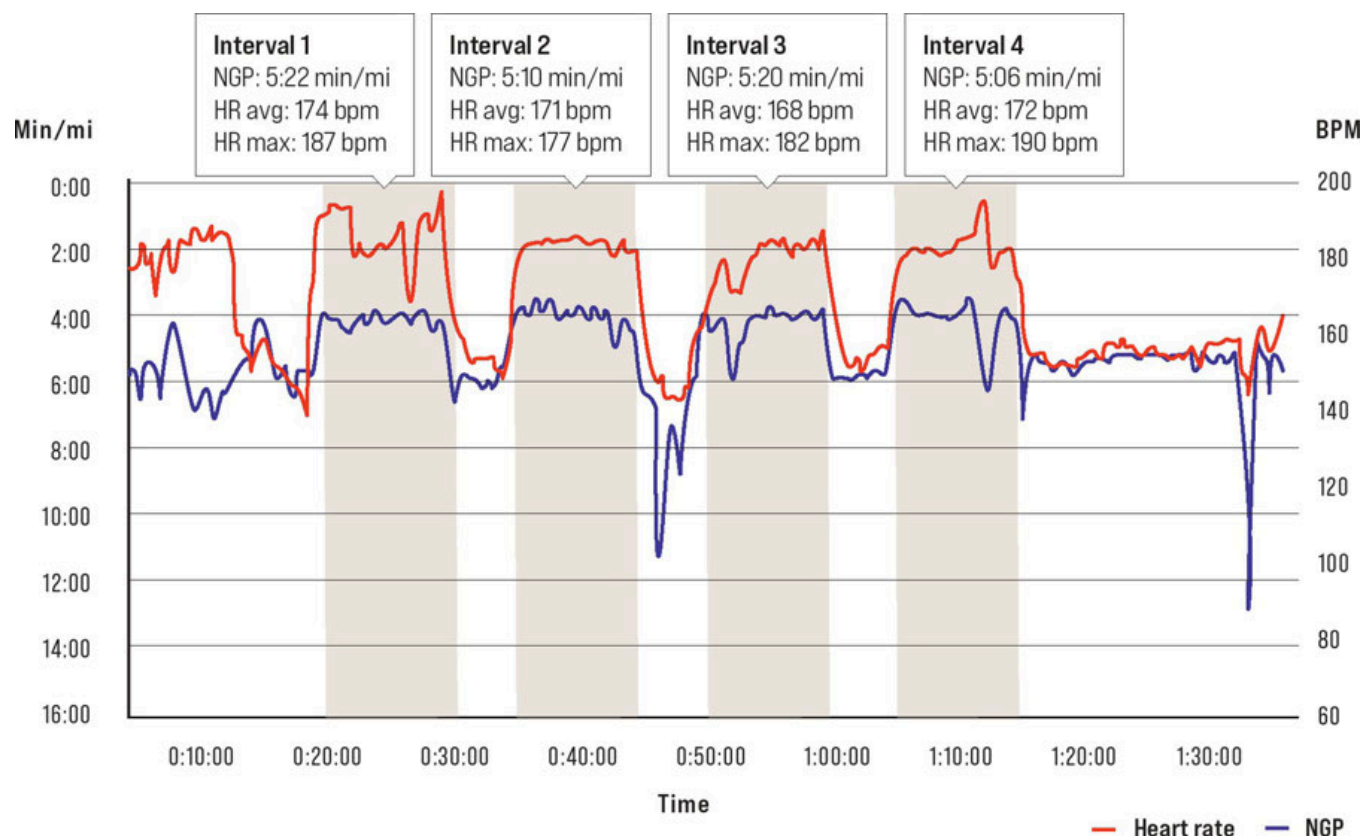


FIGURE 6.4 Two consecutive days of TemporoRun intervals. While the normalized paces are similar, the heart rate is generally depressed on the second day. Had the athlete been training using heart rate, he either would not have been able to do the workout or would have pushed too hard.

Of all the factors that affect heart rate, fatigue can get an athlete in the most trouble. When heart rate response is exaggerated and heart rates are higher than expected, athletes who are training or competing by heart rate tend to slow down. If your ranges say to run at 150 beats per minute (bpm) to sustain a pace without exceeding your lactate threshold, but your heart rate response is being boosted by 5–8 bpm due to elevated core temperature and caffeine, running at 150 bpm will result in a slower actual pace. Meanwhile, when fatigue suppresses heart rate, you may push even harder in an effort to achieve your goal intensity of 150 bpm.

Figure 6.3 is from an athlete in a 100K race. The first three climbs of the race were paced perfectly at similar NGP. However, the heart rate on the first climb is markedly higher due to the athlete's initial freshness and adrenaline rush at the start of the race. The heart rate on the third climb is suppressed because of the accumulated fatigue. Had the athlete tried to target a specific heart rate on the first climb, she would have been much too slow. Had she tried to match her heart rate from the first two climbs on the third climb, she would have run far too hard, particularly that early in the race.

There are two problems with pushing harder in response to a suppressed heart rate response. The first is that a suppressed heart rate at a given effort level or pace doesn't always mean you aren't performing as much work. Athletes who are fatigued can often perform the same workout two days in a row but experience suppressed heart rate response on the second day.

Figure 6.4 shows a runner performing interval work on back-to-back days. On the second day, the athlete is able to perform the workout at a similar NGP but at a generally lower maximum and average heart rate due to the accumulated fatigue. This phenomenon plays into the second problem caused by suppressed heart rates. If you are gauging the success of your workout by your ability to run at a specific heart rate—or within a small heart rate range—and that heart rate is difficult to achieve because of fatigue, you may just push yourself harder to achieve the heart rate number you want to see. The effort feels harder than it should, but you push through anyway. If you are truly fatigued, you're just digging the hole deeper.

CARDIAC DRIFT

One of the greatest disadvantages of using heart rate alone to gauge training intensity is “cardiac drift.” Because up to 75 percent of the energy produced in muscles is lost as heat, your body has to work to dissipate that heat to keep your core temperature from rising out of control. As you exercise—especially at higher intensities—your body uses your skin much like your car uses its radiator. Heart rate increases not only to deliver oxygen to working muscles but also to direct blood to the skin so it can supply fluid for sweat and cool off through convection (provided that the ambient temperature is lower than your core temperature). The sweat is released onto the surface of the skin so it can evaporate, which carries much of this excess heat away from the body. Much of the fluid that appears as sweat on your skin was most recently part of your bloodstream. As you lose blood plasma volume to produce sweat, your heart has to pump even faster to continue delivering the same amount of oxygen to working muscles. As a result, your heart rate will increase slightly as exercise duration increases, even if you maintain the same level of effort. The impact of cardiac drift will be lower if you are able to stay well hydrated: you're replacing the fluid lost by sweating and helping to maintain a higher overall blood volume. However, no matter how diligent you are about consuming fluids, some level of cardiac drift is unavoidable during intense endurance exercise.

You can see the impact of cardiac drift in Figure 6.5. In this heart rate file from a lactate threshold interval workout, the athlete performs three intervals at roughly the same pace, but his heart rate gets progressively higher for each effort. When athletes train by heart rate alone, they are instructed to maintain the same heart rate range for each interval. Ideally this would result in efforts of equal intensity, but as a result of cardiac drift, the first interval is actually completed at a faster pace than the subsequent ones. To the athlete, heart rate seems right on target, but they don't realize that workload is actually falling, and as a result, the workout loses some of its potential effectiveness.

Why Ultrarunners Should Embrace Perceived

Exertion

As much as I embrace the role of technology in enhancing the precision of training, there's an incredibly simple measure of workload that continues to hold its own against new gadgets and software applications. Rating of perceived exertion is the ultimate in simplicity: it is nothing more than a scale of how hard you feel you are exercising. There's not one single piece of data collected, and you don't need any special equipment. All you need is a numerical scale.

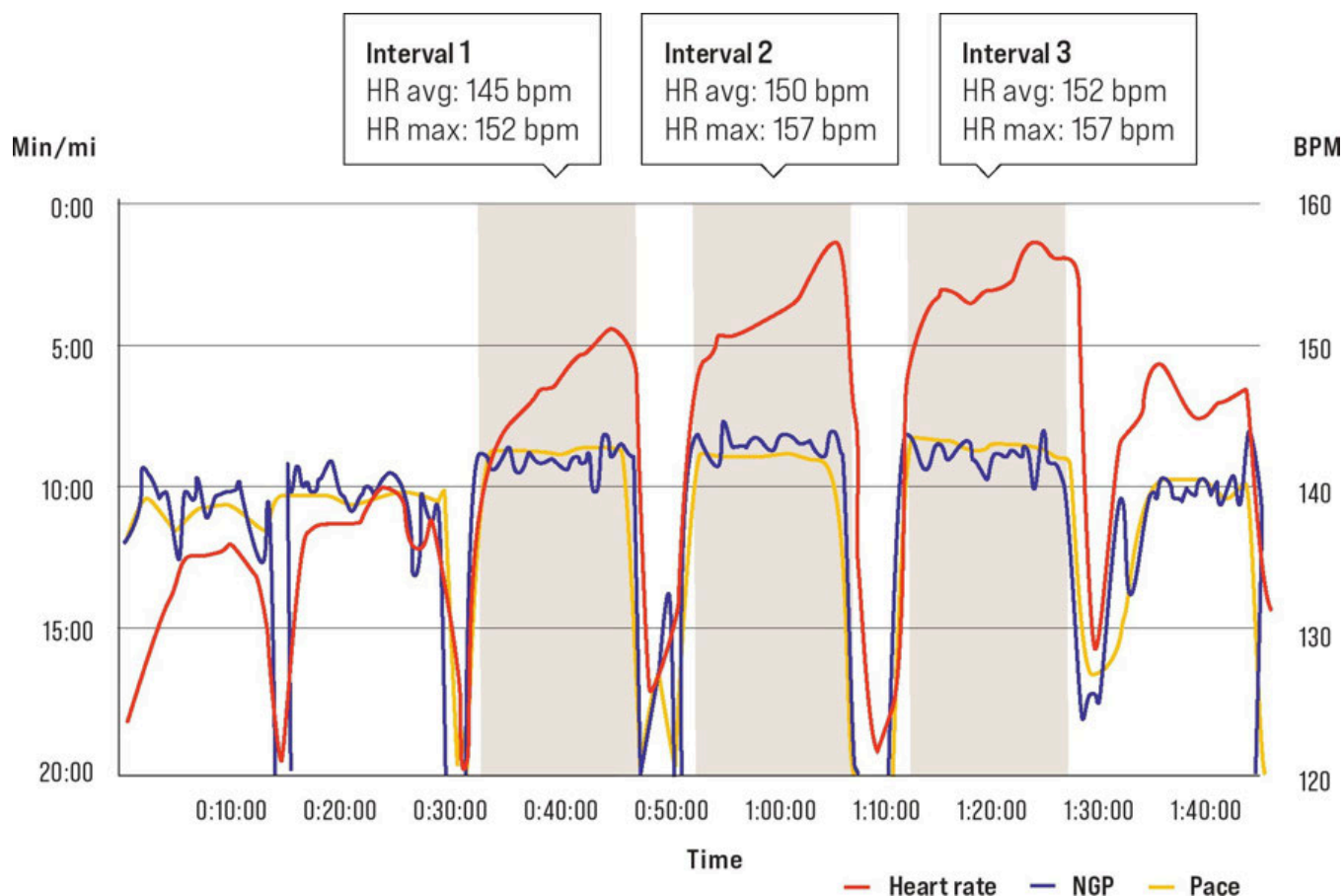


FIGURE 6.5 Impact of cardiac drift during a 3 × 10-minute TempoRun workout where the heart rate increases throughout each interval and from interval to interval.

In the physiology lab, I use the Borg Scale, which ranges from 6 to 20 (with 6 being no exertion at all and 20 being a maximum effort). Why 6 to 20? Borg's research has shown that there's a high correlation between the number an athlete chooses during exercise, multiplied by 10, and her actual heart rate at that time. In other words, if you're on a treadmill during a lactate threshold test and tell me that you feel like you're at 16, there is a pretty good chance your heart rate is around 160 bpm. This isn't absolutely true of all athletes, but you'd be surprised at how accurate the 6 to 20 scale tends to be.

Outside the lab, however, the Borg Scale isn't as helpful for athletes, most of whom find it easier to relate to a simpler 1 to 10 scale (with 1 being no exertion at all and 10 being a maximum effort). Using this scale, an endurance or "forever" pace would be a 5 or 6, a challenging aerobic pace would be a 7, medium-intensity interval work that approximates lactate threshold intensity occurs at about 8 or 9, and high-intensity intervals—which I call RunningIntervals—are the only efforts that reach 10. Just as the Borg Scale multiplies the perceived exertion number by 10 to correlate with heart rate, the number chosen on the 1 to 10 scale, multiplied by 10, seems to correlate closely to the percentage of VO₂ max that an athlete is currently maintaining.

WHAT ABOUT LACTATE THRESHOLD AND VO2 MAX TESTING?

OVER THE YEARS, I have brought most of the elite athletes I work with—and many other runners—into a physiology lab for lactate threshold and VO2 max testing. There is a sentiment that this type of testing lacks value, as VO2 max values are not useful by themselves. This is true, but physiological testing is less about finding one particular end point and more about what is happening during the continuum of intensities that are studied.

I mostly use the tests as benchmarks to establish baseline physiological values for an athlete and then build a profile of her progress through subsequent testing. On a practical level, however, I don't use the test results to create training intensity ranges for ultrarunning athletes, as I do for other types of endurance athletes. I think the most valuable thing an ultrarunner can learn from testing, and the most valuable thing I've learned from testing many ultrarunners, is that ultrarunners have lactate threshold values that represent a higher percentage of their overall VO2 max. In other words, a cyclist might come into the lab, and his lactate threshold power is 78 percent of his power at VO2 max. An elite cyclist might increase that to about 85 percent. Most marathon runners are similar. But an elite ultrarunner will frequently record a lactate threshold value at 95 to 97 percent of VO2 max!

What does that mean? For the most part it means ultrarunners have traditionally done a poor job of training to improve VO2 max and a good job of developing their physiology at lower intensities. Their training has improved performance at aerobic and lactate threshold intensities but has not had much effect on their maximum capacity to take in and deliver oxygen to working muscles. Normally I'd say having a lactate threshold at 95 to 97 percent of VO2 max is great, but in the context of ultrarunners, what it actually shows is where most ultrarunners are underdeveloped. VO2 max is your maximum potential—the ceiling or roof of the building. If you want to fit more stuff in the building (more endurance, greater performance capacity), then raise the roof!

With GPS-equipped heart rate monitors providing more detailed pacing information for runners, some athletes are tempted to relegate RPE to the trash bin of sports science history, but RPE remains critically important because it provides valuable context for the data files from a heart rate monitor or GPS watch. When you're fresh, 9:00 min/mi may feel like a moderate pace, but when you're fatigued, you may feel like you're having to work harder than normal to run that same pace. It turns out that RPE is a great earlywarning device for recognizing fatigue: your body is telling you it can still do the job but that the effort to complete it is greater.

In addition, RPE can indicate progress. For example, at the beginning of the season, a ten-mile run at 9:00 min/mi pace may feel strenuous enough to rate a 7 or even an 8. Later in the season, when your fitness has improved, running that same course at that same pace may take less out of you and feel more like a 6. To reach an RPE of 7 to 8, you may now need to hasten your pace to 8:30 min/mi.

Although many of my athletes have access to the best training technologies and gadgets on the market, I base the vast majority of their training on perceived exertion. The main reason is because everything else is irrelevant during an actual competition. When you are scrambling up a 25 percent grade in a cold thunderstorm at 10,000 feet above sea level, 65 miles into a 100-mile ultramarathon, what heart rate would define lactate threshold pace? What minute-per-mile pace should a midpack ultrarunner aim for in that scenario?

Your brain is the most valuable tool you have for monitoring and evaluating your intensity, and it's the only training tool yet designed that can determine the correct interval and racing intensities for an ultrarunner. Your job as an athlete is not to find the gadget that will give you the information you seek but, rather, to master the ability to gauge intensity and workload by perceived exertion. It's the only information you need, which is good, because it's also the only accurate information you have.

To use perceived exertion to accurately gauge your workload, you need a good understanding of what you're trying to accomplish at each intensity level, the impact that intensity level has on your body, and the ways you can detect or interpret those impacts.

RPE AND BREATHING RATE

The respiration rate in healthy individuals is highly attuned to the amount of carbon dioxide in your blood. Interestingly, in normal conditions at rest, your involuntary respiration rate is more controlled by the carbon dioxide you’re trying to get rid of than the oxygen you need to survive. As your energy demand increases and your muscles break down fuel for energy, the carbon dioxide levels in your blood increase. To keep carbon dioxide levels from rising too much, heart rate and breathing rate increase so more venous blood transports carbon dioxide to the lungs. There, pressure gradients transport carbon dioxide out of the blood (and oxygen into it), and you exhale the carbon dioxide. Table 7.1 shows how different types of workouts affect RPE and breathing rate.

Because exercise increases both your oxygen demand and your production of carbon dioxide, it makes intuitive sense that increasing exercise intensity leads to faster and deeper breathing. It’s so intuitive that it’s largely ignored, but you can monitor your breathing as a gauge of intensity. At a recovery pace (RPE 4 or 5), you should be breathing only a bit faster and deeper than when you walk at a brisk pace. If you have just finished an interval, your breathing will of course be much faster and should come down to this 4 or 5 level during your recovery period. The exception is during high-intensity interval workouts, where the recovery periods are purposely too short to allow full recovery.

The amount of carbon dioxide you produce in response to energy expenditure does not increase linearly, and this is the phenomenon athletes need to pay attention to. You produce more carbon dioxide when you burn more carbohydrate for energy. Furthermore, a higher percentage of your energy comes from carbohydrate as your exercise intensity increases because glycolysis (the partial breakdown of carbohydrate that provides energy quickly but also results in lactate production) breaks down only carbohydrate, not fat.

At rest you normally exhale less carbon dioxide than you inhale oxygen. This respiratory exchange ratio (RER; volume of CO2 expired/volume of O2 inspired) is indicative of the fuels you are burning; an RER of 0.7 to 0.85 is normal at rest or low activity levels and is associated with burning a mixture of fuels. With more intense exercise, your RER rises as you start exhaling more carbon dioxide. RER approaches and can exceed 1.0 as the rate of carbohydrate utilization increases from glycolysis.

RPE	ACTIVITY	TALKING ABILITY
1-3	Sitting on the couch	Uninhibited
4-5	Easy run	Story time!
5-6	Normal run	Comfortable conversation
7-8	Hard workout	2-3 sentences at a time
8-9	Very hard workout	5-7 words at a time
9-10	Extremely hard workout	Single word, probably four letters

TABLE 6.4 RPE and the Talk Test.

RPE AND THE TALK TEST

HOW FAST YOU'RE BREATHING impacts how easily and comfortably you can speak. I call recovery pace “story time” because you should be able to tell your training partner all about last weekend’s epic adventure without pausing. When you bring the pace up to endurance or “forever” speed, you should be able to have a comfortable conversation, but you’re probably not talking for very long before having to pause. As intensities increase, your ability to converse will dwindle to two or three sentences before you need to focus full-time on breathing. When you are running at or near your lactate threshold pace, you should only be able to say one complete sentence, maybe five to seven words. If someone tells you they are running at lactate threshold yet they are carrying on a normal conversation, they are lying to you (or they don’t know any better). Finally, there’s no talking at intensities near VO2 max, unless you count four-letter expletives and grunts.

Because higher levels of carbon dioxide drive you to breathe faster and deeper to get rid of it, and a dramatic increase in carbohydrate utilization leads to a big increase in carbon dioxide levels in the blood, your breathing rate can give you a good (not perfect) indication of when you have reached lactate threshold.

As you progress from recovery pace to an endurance or “forever” pace (RPE 5 or 6), your breathing becomes deep and rhythmic; it is faster than at recovery pace but not labored. When you step up the intensity to target the high end of your aerobic range (RPE 7), your breathing will remain deep, but you should start to feel that you are laboring to breathe fast enough. The next intensity level is the tricky one: lactate threshold (RPE 8 or 9).

The easiest way to use your breathing to find lactate threshold is to go beyond it. When you exceed lactate threshold intensity, your breathing will go from deep, labored, and in control to short and rapid. The intensity or pace you want to hold during a lactate threshold interval is below the point at which your breathing shifts from controlled to short and rapid. In contrast, when you are doing RunningIntervals (RPE 10), you want to exceed this ventilatory threshold; your breathing should be short and rapid.

WHAT ABOUT RUNNING POWER METERS?

In the early 2000s as I was ramping up my coaching career, I had the good fortune to be privy to the power meter revolution that took over cycling. The concept of cycling power meters was simple. A cycling power meter works by measuring the force being produced either on the crank arms, pedals, or hub. By measuring the force, as well as the speed of what you are measuring (the pedals, cranks, or hub), it’s a quick calculation to mechanical power ($\text{power} = \text{force} \times \text{velocity}$).

Early power meter models were expensive, clunky, and heavy—all of which made power meters rare outside of the small circle of the best Grand Tour and track riders. Yet, they were a powerful-enough tool that many of the top riders in the world used them to inform training. Fast-forward a few years to the mid- and late 2000s, and these devices became more affordable, more streamlined, and lighter. Those innovations paved the way to widespread adoption not only by elite athletes but by everyday cyclists. This revered tool, once reserved only for training, could now also be used for racing to glean valuable insights into the demands of the sport.

It is in no way a stretch to say that this transition revolutionized the way we approached training cyclists. As coaches, we had a direct way of measuring intensity and race demands as well as a simple way of transferring and analyzing all of the information. Coaches subsequently had to “learn” how to read, analyze, and aggregate power meter files to better inform their coaching practice, and I am proud to say that I helped lead some of the first coaches to be fluent in this new language. I was also present to witness power meter failures, ranging from misinterpretation of the data to design flaws of the devices (and sometimes both). All of this experience continues to shape how I work with data to better coach athletes.

Jump to present time. Companies like Stryd, RunScribe, and Polar, among others, have recently attempted to replicate the cycling power meter revolution in running. It’s a novel attempt, but in my opinion it will ultimately either fail or at least take on a different meaning. While cycling power meters directly measure the force exerted

on the crank arms, pedals, or hub, running power meters have had to interpret power through a few degrees of separation. Some companies use the movement of the foot to determine the mechanical work required for the activity and thus calculate mechanical power through a proprietary algorithm. Others take a different approach altogether by reverse engineering the metabolic power (oxygen cost) required to run. Throughout it all, the key thing to keep in mind with a running power meter is that the devices are deriving power using various algorithms, none of which is a good surrogate for directly measuring force and speed like a cycling power meter. The differences are so great that even the phrase “running power” is essentially meaningless, both in terms of its biomechanical interpretation as well as its practical use with athletes.

So, where does this leave the average trail runner when it comes to “running power?” First off, in a trail running application, determining running power is problematic because the surface cannot be accounted for (similar to the NGP issue discussed earlier). Your running power meter does not know if you are running on pavement, sand, or an alpine ridge. Therefore, regardless of how good the algorithms are or if the running power meter is providing information about your mechanical or metabolic power, interpreting the data as an intensity surrogate is a meaningless exercise. Using running power meters that measure movement at the level of the foot might have utility to inform trail runners on effective training technique or form by analyzing the precise movement of the foot over the course of a run. But as a means of determining intensity, the promise of running power meters has yet to prove its worth.

HEART RATE VARIABILITY (HRV)

LIKE MANY PEOPLE, it’s possible you got into running because it’s simple. All you need is a pair of shoes, maybe a watch, and you’re out the door. Despite that aspiration, the running world is saturated with gadgets and measurements that promise to make you better, faster, fitter, and stronger. Enter heart rate variability (HRV).

HRV is the measure of inter-heartbeat variation for a given period of time, reflected by what is known as the R-R interval (see Figure 6.6). What I mean by this is that even though your heart might beat at a rate of sixty beats per minute, those beats do not happen perfectly every second on the second. There is variability, and that variability of the timing within each heartbeat is both good and important. Measuring this variability gives us a noninvasive window into how well your autonomic nervous system is functioning. The autonomic nervous system is your body’s regulator, with the important job of helping to maintain homeostasis. When you are fatigued, your autonomic nervous system cannot adjust to internal and external stimuli as efficiently, and that is reflected in your inter-heartbeat variation being more regular (which in this case is negative).

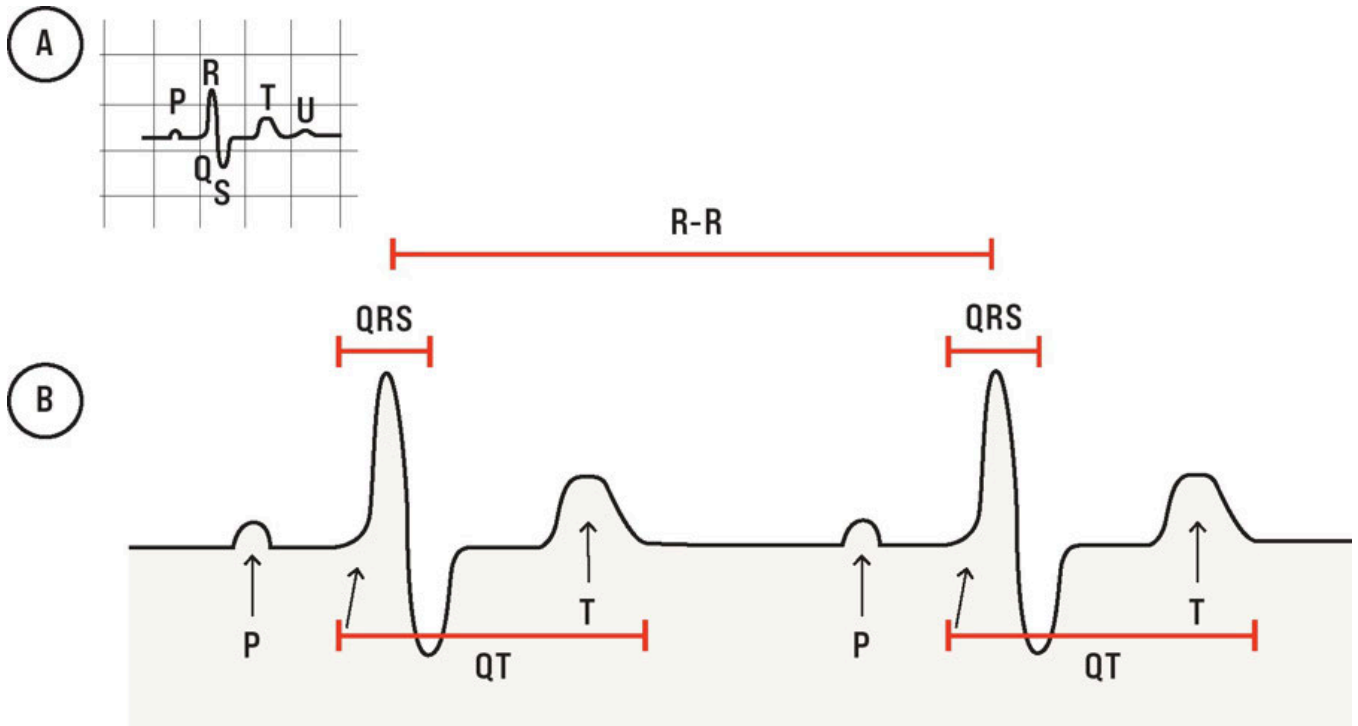


FIGURE 6.6 An EKG reading of the R-R interval across two heartbeats. Adapted from Dong (2016).

In this line of reasoning, we can utilize HRV to monitor training load, fatigue, and potentially how well you are adapting to training. Negative adaptations to long-term training would then be associated with reductions in HRV, and an increase in fitness (positive adaptation) would be associated with increases in HRV indices (Dong 2016).

What does that all mean? In order to utilize HRV, you need to do a few things:

- Establish a baseline. HRV does not provide valuable information with one singular reading but instead carries the most weight when you can monitor positive and negative trends. As with many other metrics in exercise physiology for performance, context is key.
- Take effective readings. Aside from daily readings, it's important to also utilize the best measurements possible. This means taking the readings upon waking first thing in the morning and opting for longer measurements than the default sixty-second options most applications will give you. Clinicians opt for five-minute readings. If you can fit that into your morning routine, five minutes gives you the likelihood of getting cleaner recordings.
- Understand HRV is not cut-and-dried. Like any training metric, HRV is not perfect. An increase in your HRV is not always positive, and a decrease is not always negative. Peaking for an event with a taper, an artificial rebound value forty-eight hours after a hard effort, or an overworked immune system while you fight off illness can send your daily HRV readings into uncharted territory. Therefore, it's important to utilize context and monitor trends over time; if you start using HRV, commit to gathering data for at least four weeks before making big decisions.

SUBJECTIVE FEEDBACK

Out of all of the tools I use on a day-to-day basis with my athletes, subjective feedback is the most important. Descriptions of how you felt during a workout should be incorporated into your training log, and a premium should be placed on capturing this information consistently. You can record whatever you wish, but at a minimum write down how you felt (was the workout hard, easy, did you feel particularly fast, etc.), any challenges you encountered along the way, and any significant nutrition strategies you used. You will end up using the subjective feedback to help decide when to take a recovery phase (see Chapter 10) and to determine your nutrition plan (see Chapter 14), as well as to illuminate patterns and trends in your training. Finally, deliberately capturing subjective feedback makes you more aware during the run, which in turn can drive mental and psychological development that is also critical to success (more on this in Chapter 15).

What the Coach Uses

I will let you in on a secret: very little, if any, of this information is actionable by itself. In all my years as a coach, I cannot recall a time when I have taken one piece of such information and used it to decide what an athlete should do. Rather, action is determined by the aggregate of the information, combined with feedback from the athlete. From a practical standpoint, I utilize the following system to drive the creation of and adjustments to an athlete's training. I encourage you to do something similar.

- Use TrainingPeaks to schedule workouts (see the short-range plan, as described in Chapter 12).
- Use NGP, rTSS, and Strava segments to evaluate and compare workouts.
- Use rTSS/CTL/ATL to track training load.
- Use subjective feedback from the athlete to further gauge fitness, fatigue, and motivation.
- Use knowledge and experience to synthesize the information and drive action.

No Magic Bullet

Unfortunately for trail runners, there is no magic bullet for tracking workload or work rate or for monitoring progress. There is an amalgamation of approaches you can use to get a better handle on these aspects, but no single one will give you a definitive answer. Even when better tools emerge, coaches and athletes will continue to take their training and analysis beyond the realm of what the data can provide. Soon enough, running workload and work rate will become the norm. These will replace pace, NGP, and GAP just as those metrics have largely replaced the heart rate monitor. Even when that happens, don't expect to see stoplight answers. The training picture will never be green or red, nor will it be encapsulated by my quiver of highlighters. Instead, it will be a

blur of many different colors, altogether subject to interpretation. As technology progresses, expect to be able to train more precisely and with greater knowledge—but don't expect the information to do the training for you.

Many of the training file graphs produced throughout this book were done through TrainingPeaks' WKO5 software. I would like to thank Frank Pipp and Mike Bonenberger at TrainingPeaks for their assistance.

KEY POINTS ON TOOLS TO TRACK YOUR TRAINING

GPS watches are going to be the primary tool to track your training.

- **Choose the most accurate settings when recording intervals.**

Tools like TrainingPeaks, Strava, and Final Surge can harness the information collected on your watch and make it informative and actionable.

Track volume by time, not miles.

TSS can be used in concert with volume to inform you about training load, fitness, and fatigue.

- **Realize the limitations of TSS, and don't let it dictate your training.**

Heart rate is not a good tool for determining intensity. You can use it after the fact in training analysis as long as you know what factors confound it.

RPE should be used to calibrate intensity during workouts and races.

Subjective feedback is also critical to capture during day-to-day training.

Synthesize all of the information available to drive training.

